

Project Executable Files

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes (README.md)
3	requirements.txt / package.json / dependency file	Yes (requirements.txt, requirements-dev.txt)
4	Database schema / seed files (if applicable)	NA (no database)
5	Environment configuration file (.env.example or similar)	Yes (.env.example)
6	Deployed application URL (if hosted)	NA (runs locally)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA (web application)
8	Dockerfile / Containerization config (if applicable)	Yes (Dockerfile, .dockerignore)
9	Test files and test results	Yes (tests/); results in 6.Project Testing/evidence/ once the team has run them
10	Demo video or walkthrough (if required)	https://drive.google.com/file/d/1ipvU__y_tIdTWoixNRHC2KDrYWURZdPI/view?usp=drive_link

Step 2: File / Folder Structure

EduGenie/	
-- main.py	FastAPI app: routes, validation, error handling
-- gemini_utils.py	Gemini service: API key, model fallback, safe errors
-- config.py	Settings from environment variables
-- models.py	Pydantic request schema
-- qna.py	/qa - question answering with follow-up context
-- explanation_module.py	/explain - structured concept explanation
-- quiz_module.py	/quiz - 3 MCQs as JSON, cleaned and validated
-- summary_module.py	/summarize
-- learning_path.py	/learn/recommendations
-- check_gemini.py	Activity 1.3: Gemini connectivity check
-- requirements.txt	Runtime dependencies
-- requirements-dev.txt	Test and documentation tooling

```

|-- pytest.ini           Test configuration
|-- .env.example         Environment template (copy to .env)
|-- Dockerfile, .dockerignore  Container build
|-- templates/           index.html
|-- static/              style.css, app.js
|-- tests/               test_app.py, conftest.py, perf_load.py
|-- tools/               build_docs.py, capture_screenshots.py
|-- docs-src/            Editable Markdown sources and project.json for the PDFs
|-- 1. Brainstorming & Ideation/ ... 8.Project Demonstration/   Phase deliverables (PDF)
`-- README.md

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not hosted - runs locally at http://localhost:8000 (see run instructions)
Login Credentials (Demo)	Not needed. EduGenie has no accounts. A Gemini API key is required in <code>.env</code>
Platform / Hosting Provider	Local Uvicorn server; Docker image available for Render, Railway, AWS or Azure
Repository Link	https://github.com/Dhanam0706/EduGenie
Demo Video Link	https://drive.google.com/file/d/1ipvU__y_tIdTWoixNRHC2KDrYWURZdPI/view?usp=drive_link

Step 4: Run Instructions

1. Install Python 3.10 or newer and Git.
2. `git clone https://github.com/Dhanam0706/EduGenie` then `cd EduGenie`
3. Create and activate a virtual environment: `python -m venv venv` then `venv\Scripts\activate` (Windows) or `source venv/bin/activate` (macOS/Linux).
4. `pip install -r requirements.txt`
5. Copy `.env.example` to `.env` and put your Gemini key from <https://aistudio.google.com/apikey> in `GEMINI_API_KEY`.
6. Optional check: `python check_gemini.py` (confirms Gemini access and shows which model works).
7. Start the server: `uvicorn main:app --reload` (or `python main.py`).
8. Open <http://localhost:8000> and try each task. Interactive API docs are at <http://localhost:8000/docs>.
9. Run the tests: `pip install -r requirements-dev.txt` then `pytest tests`.
10. Docker alternative: `docker build -t edugenie .` then `docker run -p 8000:8000 --env-file .env edugenie`.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	AI answers can contain mistakes	Students should check important facts against their textbook; this is stated in the demo
2	A Gemini API key is required, and the free tier has usage limits	Create a free key in Google AI Studio; wait a moment if a "too many requests" message appears
3	Gemini model names change over time	Set <code>GEMINI_MODEL</code> in <code>.env</code> ; the app also tries fallback models automatically
4	Follow-up context lives only in the open browser tab	Reloading the page starts a new conversation; saved history is planned

S.No	Known Issue / Limitation	Workaround / Status
5	Text typed by the student is sent to the Google Gemini API	Do not enter private information